SEMANTICA ÎN LOGICA PROPOZIȚIONALĂ

Ex 1. : $S = \{A_1, A_2\}$

$F: S \to \{a, f\}$, $F(A_1) = a$, $F(A_2) = f$

$V_F[(A_1 \vee A_2) \to A_2]$

$= V_F(A_1 \vee A_2) \rightsquigarrow V_F(A_2) = [V_F(A_1) \cup V_F(A_2)] \rightsquigarrow V_F(A_2)$

$= [a \cup f] \rightsquigarrow f = a \rightsquigarrow f = \boxed{f}$

$F(A_1) = a$, $F(A_2) = a$

$V_F[(A_1 \vee A_2) \to A_2) = [a \cup a] \rightsquigarrow a = a \rightsquigarrow a = \boxed{a}$

Ex 2: a). $A \vee \neg A$

I. $F(A) = a$

$V_F(A \vee \neg A) = V_F(A) \cup V_F(\neg A) = V_F(A) \cup [\sim V_F(A)] =$

$= F(A) \cup \sim F(A) = a \cup \sim a = a \cup f = \boxed{a}$

II. $F(A) = f$

$V_F(A \vee \neg A) = f \cup (\sim f) = f \cup a = \boxed{a}$

$\Downarrow$ $A \vee \neg A$ = tautologie

b). $[(A \to B) \to A] \to A$

I. $F(A) = a$, $F(B) = a$

$V_F\{[\underbrace{(A \to B) \to A}_{a}] \to \underbrace{A}_{a}\} = V_F[\underbrace{(A \to B)}_{a} \to \underbrace{A}_{a}] \rightsquigarrow V_F(A)$

$= [V_F(A \to B) \rightsquigarrow V_F(A)] \rightsquigarrow V_F(A) =$

$= \{[(V_F(A) \rightsquigarrow V_F(B)] \rightsquigarrow V_F(A)\} \rightsquigarrow V_F(A)$